

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
<https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1>
3. Khan Academy, Compound Circuits:
<https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power>
4. EEVBlog, Voltage Dividers:
https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWIH0UOZNtOn-FIFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
<https://www.youtube.com/watch?v=WBfAEeEzDlq&list=PLvOlSehNtuHtWIH0UOZNtOn-FIFCn1GYw&index=13>

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

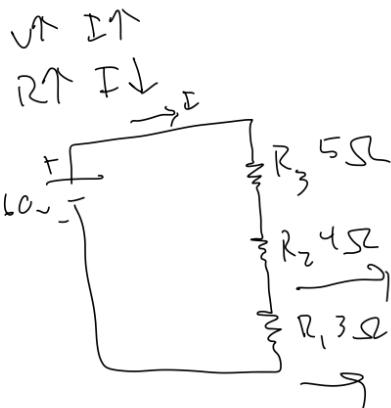
1. MIT OpenCourseWare Power Electronics Course:
 - a. [Lecture 5: Intro to DC/DC, Part 1](#)
2. [Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic](#)
3. [Power Electronics by Daniel W. Hart](#)

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Power electronics learning notes:

Ohms Law (organic chemistry tutor)

Ohms Law: $V = IR$
 $\uparrow \quad \uparrow \quad \leftarrow$
 volts amps ohms



Sum of voltage drops = battery voltage

$$V_1 = I R_1$$

$$15V = 5 \cdot 3$$

$$V_2 = I R_2$$

$$5 \cdot 4 = 20V$$

$$V_3 = I R_3$$

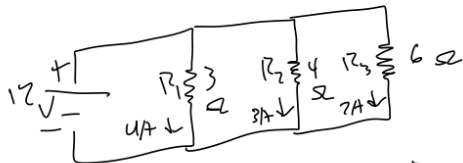
$$5 \cdot 5 = 25V$$

$$15 + 20 + 25 = 60V!$$

$$\frac{1}{3} + \frac{1}{4} = \frac{7}{12} + \frac{1}{6} = \frac{9}{12} = \frac{4}{3}$$

$$12 = \frac{4}{3} I$$

$$9 = I$$



$$V_1 = I_1 R_1$$

$$12 = 3 I_1$$

$$I_1 = 4A$$

$$V = I_2 R_2$$

$$12 = 4 I_2$$

$$I_2 = 3A$$

$$V = I_3 R_3$$

$$12 = 6 I_3$$

$$I_3 = 2A$$

$$4 + 3 + 2 = 9$$

$$R \uparrow I \downarrow$$

$$R \downarrow I \uparrow$$



$$R_T = R_1 + R_2 + R_3$$

$$R_T = 3 + 4 + 5 = 12$$

$$60V = 12 I$$

$$I = 5 \text{ amps}$$

$$12 = 4 I$$

$$I = 3 \text{ amps}$$

Khan Academy video:

(Volts)

Voltage (V) - Electric Potential

(Amps)

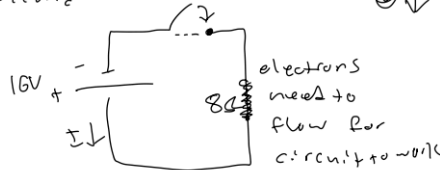
PE per unit charge

current (I) - how much

charge per unit time charge flowing

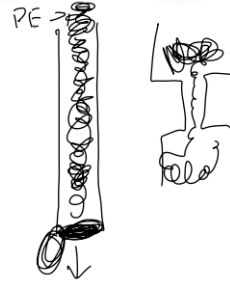
Resistance (R) (ohms) in a second

$V = IR$ how much
charge flow
impeded

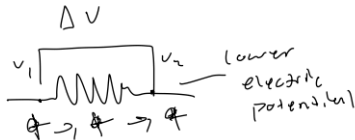


$$10 = 8I$$

$$I = 2 \text{ Amps}$$



Second Khan Academy Vid:



$$V = \frac{U}{q}$$

more PE than ending with
decreasing electrons potential
energy. speed is not increasing
as current must be the same on
both sides of the resistor. Energy
is lost to heat, heating up the
resistor as PE decreases

$$P = I^2 R \text{ also work!}$$

$$P = \frac{\Delta V^2}{R}$$

$$U_e = qV$$

$$qV_2 - qV_1 = \Delta U_e$$

$$P = \frac{E}{t} = \frac{\Delta U_e}{t} = q \frac{(V_2 - V_1)}{t}$$

$$= \frac{q \Delta V}{t}$$

$$P = \left(\frac{q}{t}\right) \Delta V$$

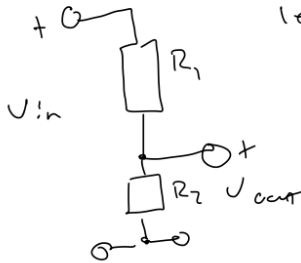
$$= I \Delta V \text{ watts}$$

$$(J/s)$$

works for
anything that turns
electric potential energy
into some other kind
of energy

Voltage Dividers; Used to get a known precise fraction of things to use in analog circuits

V_{out} must be less than V_{in}



$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{IN}$$

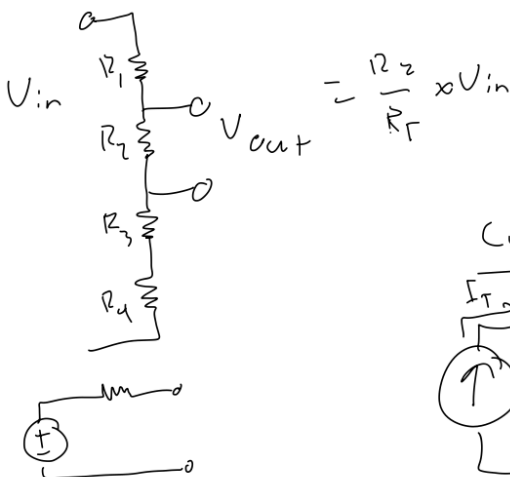
$$R_s = \sum R$$

$$\frac{1}{R_p} = \sum \frac{1}{R}$$

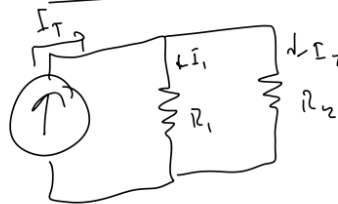
(for parallel)

Rules of thumb;

- For 10% lower cost use x10 value (look res in 1% x100 (1000k with 10k) parallel with 10k)
- 1% x1000 (10000k)
- $\frac{1}{2}$ 2 resistors in parallel // (same value)
- $\frac{1}{3}$ 3 ///



Current Dividers



$$I_1 = \frac{R_2}{R_1 + R_2} \times I_T$$

only for 2 resistors

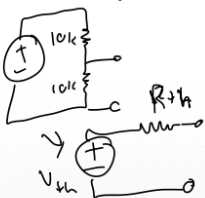
Voltage source

Current source

- Thevenin equivalent

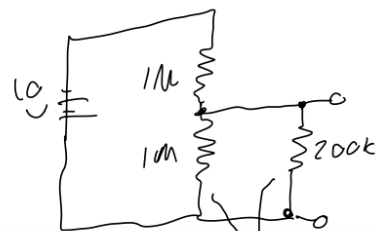
$$V_{th} = V_{open}$$

$$R_{th} = \frac{V_{open}}{I_{short}}$$



$$I_N = \frac{V_{th}}{R_{th}}$$

$$R_N = R_{th}$$



$$\frac{1}{\frac{1}{200k} + \frac{1}{1M}} = 166.6k$$

make it into one resistor

$$V_{th} = \frac{R_2}{R_1 + R_2} \times 10V = 5V$$

$$R_{th} = \frac{V_{th}}{I_{sk}} = \frac{5V}{1mA} = 5k$$

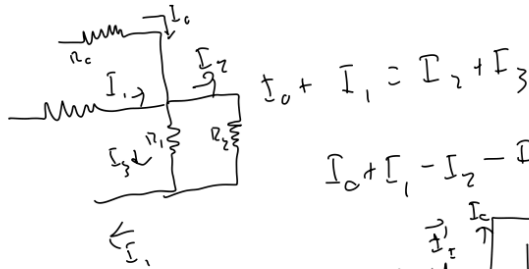
$$\frac{166.6k}{1M + 166.6k} \times 10V = 1.428V$$

Kirchoff's Laws:

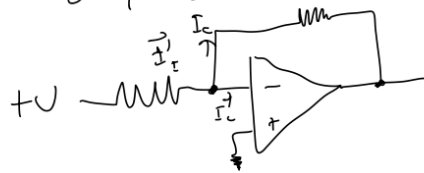
Current Law:

Sum of currents into a junction

= Sum of currents out of junction



$$I_0 + I_1 - I_2 - I_3 = 0$$



$$I_i = I_o + I_c$$

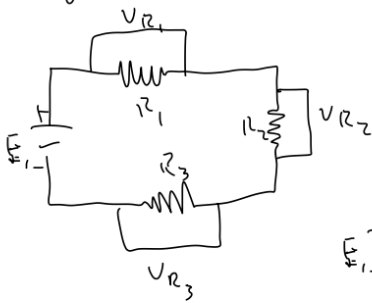
Voltage Law:

Sum of applied Voltages

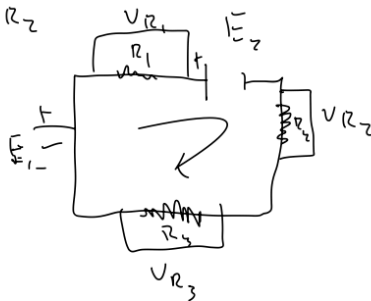
around a loop = Sum of voltage

drops around that loop

- to + for voltage
+ to - = - source



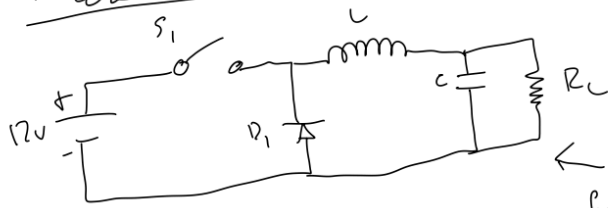
$$E_1 = V_{R1} + V_{R2} + V_{R3}$$



$$E_1 - E_2 = V_{R1} + V_{R2} + V_{R3}$$

$$E_1 = V_{R1} + V_{R2} + V_{R3} + E_2$$

Back converter circuit:



← this circuit follows V_L as $I \uparrow$ instead of losing energy to heat

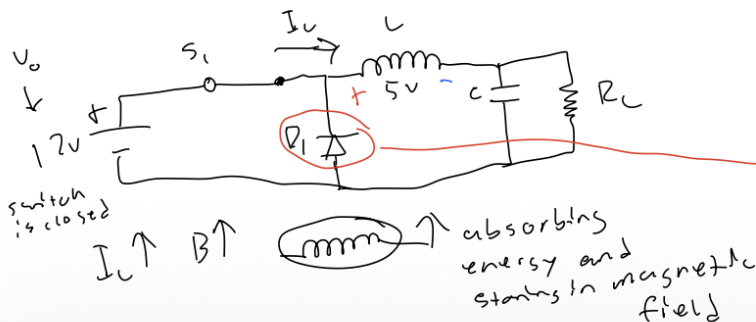
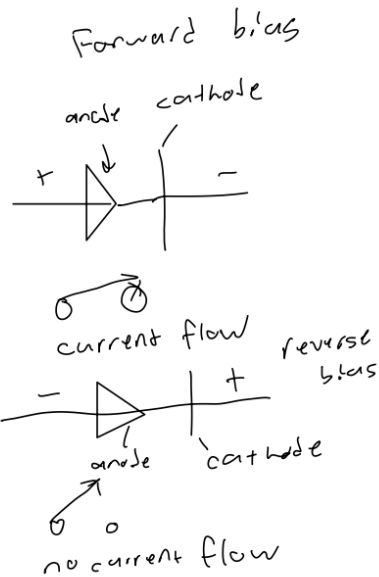
Inductor: is a passive electronic component which stores electrical energy in a magnetic field when electric current flows through it and opposes sudden changes in that current, measured in Henry (SH)

In DC circuit, once current stabilizes, inductor acts like normal wire

Diode: is a two-terminal semiconductor component that allows electric current to only flow in one direction

Forward bias: current flows freely once voltage overcomes a small threshold

Reverse bias: diode acts as an insulator and blocks current flow



It is in reverse bias so current cannot flow against it

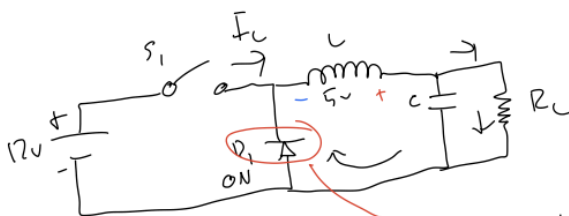
$$V_L = V_o - V_c$$

$$V_L = V_{out} - V_{in}$$

$$V_o - V_c = 12 - 5 = 7V$$

how does current $\uparrow$?

when current flows from battery through L , it also charges the capacitor and current flows through the resistor



Switch is open

$$I_L \downarrow \quad B \downarrow$$

S_1 is closed

$R_L \rightarrow I_B$ (current from battery)

$$I_B + I_L > \frac{I_B}{\tau}$$

S_1 is open

$R_L \rightarrow I_L$ (current from inductor)

if $V_{inductor} > V_{capacitor}$

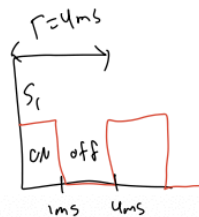
it can charge capacitor.

if equal they discharge through the load resistor

How to calculate output voltage?

$$V_{out} = V_{in}(D)$$

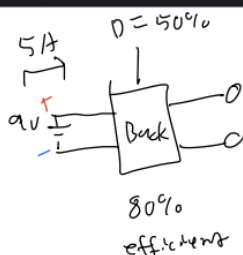
$$D(\text{duty cycle}) = \frac{t_{on}}{T}$$



$$D = \frac{1}{4} = 25\%$$

duty cycle

$$V_{out} = 12V(0.25) = 3V$$



$$P_{in} = IV$$

$$P_{in} = 45W$$

$$P_{out} = 45(0.8) = 36$$

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{36W}{4.5V} = 8A$$

$$V_{out} = 9(0.5) = 4.5V$$